

Basic Inequalities in Olympiad Problems

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June 21, 2025

Here are a few conventions and notations used throughout the passage.

- $\sum_{\text{cyc}} f(a_1, a_2, \dots, a_n)$ refers to cycling through all n summands

$$f(a_1, a_2, \dots, a_n), f(a_2, a_3, \dots, a_1), \dots, f(a_n, a_1, \dots, a_{n-1})$$

- $\sum_{\text{sym}} f(a_1, a_2, \dots, a_n)$ cycles through all $n!$ permutations in a similar manner.
- Unless specified otherwise, all inequalities will be for positive reals.

This document will attempt to demonstrate how we can tackle most Olympiad problems with a handful of inequalities (Of course, one can think of techniques like SOS and Chinese Dumbass Notation as extensions of AM-GM, but I won't do so here). First, let's consider the backbone of olympiad inequalities, the AM-GM-HM inequality.

1 AM-GM-HM

Theorem 1:

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n} \geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}$$

Note

In general the function

$$f(x) = \left(\frac{a_1^x + a_2^x + \dots + a_n^x}{n} \right)^{\frac{1}{x}}$$

is non-decreasing

1.1 Practice Problems

- 1 Find the minimum of $2^{3x^2-2x+1} + 2^{2x-3x^2+3}$.
- 2 Prove that $(a+b)(b+c)(c+a) \geq 8abc$.
- 3 Prove that $a^4 + b^4 + c^4 \geq abc(a+b+c)$.

2 Cauchy-Schwarz

Additionally, we will need the Cauchy-Schwarz inequality, which states that

Theorem 2:

$$(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2) \geq (a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2$$

2.1 Practice Problems

1 Prove Titu's lemma, which states that

$$\sum_{\text{cyc}} \frac{a_i^2}{b_i} \geq \frac{(a_1 + a_2 + \cdots + a_n)^2}{b_1 + b_2 + \cdots + b_n}$$

2 Prove that if $abc = 1$, then $a^2 + b^2 + c^2 \geq a + b + c$

3 Prove Nesbitt's inequality, which states that

$$\sum_{\text{cyc}} \frac{a}{b+c} \geq \frac{3}{2}$$

3 Examples

Here are some examples of these theorems in Olympiad problems, in increasing order of difficulty.

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Example 1: Prove that if $a + b + c = 1$, then

$$\frac{a}{a^2 + b^3 + c^3} + \frac{b}{b^2 + a^3 + c^3} + \frac{c}{c^2 + a^3 + b^3} \leq \frac{1}{5abc}$$

Proof:

Homogenising, $\sum_{\text{cyc}} \frac{5a^2bc}{a^2+b^3+c^3} = \sum_{\text{cyc}} \frac{5a^2bc}{a^3+b^3+c^3+a^2b+a^2c}$. Finally,

$$\sum_{\text{cyc}} \frac{5a^2bc}{a^3 + b^3 + c^3 + a^2b + a^2c} \stackrel{AM-GM}{\leq} \sum_{\text{cyc}} \frac{5abc}{3bc + ab + ac} = \sum_{\text{cyc}} \frac{5}{\frac{1}{a} + \frac{1}{a} + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}} \stackrel{AM-HM}{\leq} \sum_{\text{cyc}} \frac{3a + b + c}{5} = 1$$

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Example 2: If $a(a^3 + b^3) = b(b^3 + c^3) = c(c^3 + a^3)$, Prove that $a = b = c$.

Proof: Notice that $\frac{a^3}{c} + \frac{b^3}{c} = a^2 + \frac{c^3}{a}$

Summing cyclically gives

$$\frac{a^3}{c} + \frac{b^3}{a} + \frac{c^3}{b} = a^2 + b^2 + c^2$$

However

$$\frac{a^3}{c} + \frac{b^3}{a} + \frac{c^3}{b} = \frac{a^4}{ac} + \frac{b^4}{ab} + \frac{c^4}{bc} \geq \frac{a^2 + b^2 + c^2}{ab + bc + ca} (a^2 + b^2 + c^2) \geq a^2 + b^2 + c^2$$

Since $(a - b)^2 = (b - c)^2 = (c - a)^2 = 0$, $a = b = c$.

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Example 3: Prove that if $xyz = 1$, then

$$\left(x - 1 + \frac{1}{y}\right) \left(y - 1 + \frac{1}{z}\right) \left(z - 1 + \frac{1}{x}\right) \leq 1$$

Proof: Let $x = \frac{a}{b}$, $y = \frac{b}{c}$ and $z = \frac{c}{a}$.

The inequality reduces to

$$(a + b - c)(a - b + c)(-a + b + c) \leq abc$$

WLOG let $a = \max(a, b, c)$

If $a \geq b + c$, then the LHS is positive but the RHS is negative, and we are done.

So, assume $a + b > c$, $b + c > a$, $a + c > b$ so let $a = x_0 + y_0$, $b = y_0 + z_0$, $c = z_0 + x_0$ by Ravi substitution (Think of this as constructing an incircle, and notice that the lengths of the two tangents from a point to the incircle are equal).

We need $8x_0y_0z_0 \leq (x_0 + y_0)(y_0 + z_0)(z_0 + x_0)$, which follows by AM-GM on $x_0 + y_0$, $y_0 + z_0$, $z_0 + x_0$.

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Example 4: Prove that if $xyz \geq 1$, then

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0$$

Proof: This is equivalent to

$$1 - \frac{x^5 - x^2}{x^5 + y^2 + z^2} + 1 - \frac{y^5 - y^2}{x^2 + y^5 + z^2} + 1 - \frac{z^5 - z^2}{x^2 + y^2 + z^5} \leq 3$$

$$\iff (x^2 + y^2 + z^2) \cdot \sum_{\text{cyc}} \frac{1}{x^5 + y^2 + z^2} \leq 3$$

By Cauchy,

$$(x^5 + y^2 + z^2)(yz + y^2 + z^2) \geq (\sqrt{x^5 yz} + y^2 + z^2)^2 \geq (x^2 + y^2 + z^2)^2$$

So,

$$(x^2 + y^2 + z^2) \cdot \sum_{\text{cyc}} \frac{1}{x^5 + y^2 + z^2} \leq (x^2 + y^2 + z^2) \sum_{\text{cyc}} \frac{y^2 + z^2 + yz}{(x^2 + y^2 + z^2)^2} = \frac{2(x^2 + y^2 + z^2) + xy + yz + zx}{x^2 + y^2 + z^2} \leq 3$$

4 Problems

4.1 Easy

- 1 If the non-negative reals x, y, z satisfy $x^2 + y^2 + z^2 = x + y + z$. Prove that

$$\frac{x+1}{\sqrt{x^5+x+1}} + \frac{y+1}{\sqrt{y^5+y+1}} + \frac{z+1}{\sqrt{z^5+z+1}} \geq 3.$$

- 2 Let a, b, c be positive real numbers such that $abc = \frac{2}{3}$. Prove that:

$$\frac{ab}{a+b} + \frac{bc}{b+c} + \frac{ca}{c+a} \geq \frac{a+b+c}{a^3+b^3+c^3}.$$

- 3 For all positive real numbers a, b, c , prove that

$$\frac{(a^2+bc)^2}{b+c} + \frac{(b^2+ca)^2}{c+a} + \frac{(c^2+ab)^2}{a+b} \geq \frac{2abc(a+b+c)^2}{ab+bc+ca}.$$

- 4 Let $n > k > 1$ be positive integers. Determine all positive real numbers $a_1, a_2, \dots, a_n$ which satisfy

$$\sum_{i=1}^n \sqrt{\frac{ka_i^k}{(k-1)a_i^k+1}} = \sum_{i=1}^n a_i = n.$$

- 5 Let a, b, c be positive real numbers such that

$$\frac{ab}{1+bc} + \frac{bc}{1+ca} + \frac{ca}{1+ab} = 1$$

Prove that

$$\frac{1}{a^3} + \frac{1}{b^3} + \frac{1}{c^3} \geq 6\sqrt{2}$$

- 6 Let a, b, c be positive reals such that $abc = 1$. Prove the inequality

$$\sum_{\text{cyc}} \frac{2a^2 + \frac{1}{a}}{b + \frac{1}{a} + 1} \geq 3$$

- 7 Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 + (a+b+c)^2 \leq 4$. Prove that

$$\frac{ab+1}{(a+b)^2} + \frac{bc+1}{(b+c)^2} + \frac{ca+1}{(c+a)^2} \geq 3.$$

4.2 Medium

- 1 Let x, y, z be real numbers, each greater than 1. Prove that

$$\frac{x+1}{y+1} + \frac{y+1}{z+1} + \frac{z+1}{x+1} \leq \frac{x-1}{y-1} + \frac{y-1}{z-1} + \frac{z-1}{x-1}$$

2 Prove that

$$\frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} + \frac{z^2}{(z-1)^2} \geq 1$$

for all real numbers x, y, z , each different from 1, and satisfying $xyz = 1$.

3 Let a, b, c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

4 Suppose that a, b, c, d are positive real numbers satisfying $(a+c)(b+d) = ac + bd$. Find the smallest possible value of

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}.$$

5 Let a, b, c be real numbers satisfying

$$\max\{a(b^2 + c^2), b(c^2 + a^2), c(a^2 + b^2)\} \leq 2abc + 1$$

Prove that

$$a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2) \leq 6abc + 2$$

and determine all cases of equality.

6 Let $n \geq 3$ be an integer, and let $a_2, a_3, \dots, a_n$ be positive real numbers such that $a_2 a_3 \cdots a_n = 1$. Prove that

$$(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n.$$

7 The real numbers a, b, c, d are such that $a \geq b \geq c \geq d > 0$ and $a + b + c + d = 1$. Prove that

$$(a + 2b + 3c + 4d)a^a b^b c^c d^d < 1$$

4.3 Hard

1 Let a, b, c be positive reals such that $abc = 1$. Show that

$$\frac{1}{a^5(b+2c)^2} + \frac{1}{b^5(c+2a)^2} + \frac{1}{c^5(a+2b)^2} \geq \frac{1}{3}.$$

(If you know Hölder or Jensen's then think of this as a part of the easy category)

2 Prove that for any real numbers $a, b, c, d \geq \frac{1}{3}$ the following inequality holds:

$$\sqrt{\frac{a^6}{b^4 + c^3} + \frac{b^6}{c^4 + d^3} + \frac{c^6}{d^4 + a^3} + \frac{d^6}{a^4 + b^3}} \geq \frac{a + b + c + d}{4}$$

3 Given a positive integer k , let $r_2(k)$ denote the least power of 2 not dividing k . For example, $r_2(12) = 8$ and $r_2(1) = 2$.

Find the smallest positive integer n such that the inequality

$$n \sum_{i=1}^n x_i^2 + \sum_{1 \leq i < j \leq n} r_2(j-i) x_i x_j \geq 2025$$

holds for all real numbers $x_1, x_2, \dots, x_n$ satisfying $x_1 + x_2 + \cdots + x_n = 1$.

4 Find the maximal value of

$$S = \sqrt[3]{\frac{a}{b+7}} + \sqrt[3]{\frac{b}{c+7}} + \sqrt[3]{\frac{c}{d+7}} + \sqrt[3]{\frac{d}{a+7}},$$

where a, b, c, d are nonnegative real numbers which satisfy $a + b + c + d = 100$.

5 Let a, b and c be positive real numbers satisfying $\min(a + b, b + c, c + a) > \sqrt{2}$ and $a^2 + b^2 + c^2 = 3$.
Prove that

$$\frac{a}{(b + c - a)^2} + \frac{b}{(c + a - b)^2} + \frac{c}{(a + b - c)^2} \geq \frac{3}{(abc)^2}.$$

6 a, b, c are positive reals. Prove that

$$\sqrt{\frac{a^4 + 2b^2c^2}{a^2 + 2bc}} + \sqrt{\frac{b^4 + 2c^2a^2}{b^2 + 2ca}} + \sqrt{\frac{c^4 + 2a^2b^2}{c^2 + 2ab}} \geq a + b + c$$